

ADVANCED SPREADSHEET APPLICATIONS (235) NATIONAL 2025

PRODUCTION

Job 1: Create Macro _____ (100 points)

Job 2: Conditional Formatting _____ (80 points)

Job 3: Charts _____ (100 points)

TOTAL POINTS _____ (***280 points***)

Industry Certification (50 minutes)

MOS Excel Expert Certification

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

EXAM GUIDELINES:

1. Ensure this test booklet contains Jobs 1-3.
2. Key all jobs according to the instructions given.
3. Correct any and all formatting, spelling or grammar errors. Use the formatting guide in the *Style & Reference Manual*.
4. Your name or initials should *not* appear on any work you submit. Use your Member ID in any occasion you would normally key your reference initials.
5. In the lower **right** hand corner of ALL work submitted (unless otherwise specified), key your Member ID and job number.
6. If you complete the event before the end of the time allotted, notify the proctor. Time may be considered a factor in determining a winner when there is a tie score.
7. Place your scoring sheet on top of your jobs. Jobs should be placed in numerical order.

GRADER: points are all or none unless otherwise noted!

Job 1 (Create Macro)	Points Possible	Points Earned
Member # and Job # in right section of footer	5	
Create Macro named 'Nationals Baby'	10	
Macro shortcut key: 'N'	5	
Macro description: 'This macro will format table'	5	
Macro function: Set table font to Times New Roman 12 pt	5	
Macro function: Insert 3 rows between rows 1 & 2.	5	
Macro function: Row 1 merge cells A1:H1, Height: 33 pixels, Font: 26 pt., Fill color: dark red, Text: white.	10	
Macro function: Row 2 & 4 Height: 15 pixels	5	
Macro function: Row 3 Height: 20 pixels	5	
Macro function: Row 5 Height: 20 pixels A5:H5 Font: 14 pt. Background White 1, Darker 25% Text: black	10	
Macro function: column H set to currency	5	
Macro button: Macro 'Nationals Baby' assigned, button reads 'Nationals Baby'	5	
Columns resized to accommodate ALL data	5	
Spreadsheet printed	5	
Screenshot: 'Assign Macro' printed	5	
Screenshot: 'Macro & Macro Options' printed	5	
Word document printed with Visual Basic coding for Macros	5	
Subtotal	100	
Job 2 (Conditional Formatting)	Points Possible	Points Earned
Member # and Job # in right section of footer	5	
Cells B3:C3 merged	5	
Cell BC3: text 'Select Form' and formatted same as row 5	5	
Cell D3: drop-down list created from the data in column B	25	
Cell D3: conditional formula to format all rows in the table that contain the data displayed (selected) in cell D3 with 'Light Red Fill with Dark Red Text'	30	
Columns resized to accommodate ALL data	5	
Print Data	5	
Subtotal	80	
Job 3 (Chart)	Points Possible	Points Earned
Member # and Job # in right section of footer	5	

New sheet called 'Salary by Gender & Ethnicity'	5	
Pivot Table: Filters: Sex	10	
Pivot Table: Rows: Ethnicity	10	
Pivot Table: Values: Salary & Department	20	
Column B: currency	5	
Table data filtered to display "Female" employee information	10	
Table title: 'Sheet, Salary by Gender & Ethnicity', font:14 pt., fill: White, Background 1, Darker 25%.	15	
Font is Times New Roman for all of the chart text	5	
Columns resized to accommodate ALL data	5	
Landscape Orientation	5	
Print Chart	5	
Subtotal	100	
TOTAL POINTS	280	

Job1 Print Out

Fruit Prices							
Fruit	Form	RetailPrice	RetailPriceUni	Yield	CupEquivalentSize	CupEquivalentUni	CupEquivalentPrice
Apples	Fresh	\$ 1.52	per pound	\$0.90	\$	0.24	pounds \$ 0.41
Apples, applesauce	Canned	\$ 1.07	per pound	\$1.00	\$	0.54	pounds \$ 0.58
Apples, ready-to-drink	Juice	\$ 0.78	per pint	\$1.00	\$	8.00	fluid ounces \$ 0.39
Apples, frozen concentrate	Juice	\$ 0.59	per pint	\$1.00	\$	8.00	fluid ounces \$ 0.29
Apricots	Fresh	\$ 2.97	per pound	\$0.93	\$	0.36	pounds \$ 1.16
Apricots, packed in juice	Canned	\$ 1.69	per pound	\$1.00	\$	0.54	pounds \$ 0.91
Apricots, packed in syrup or water	Canned	\$ 2.06	per pound	\$0.65	\$	0.44	pounds \$ 1.40
Apricots	Dried	\$ 6.62	per pound	\$1.00	\$	0.14	pounds \$ 0.95
Bananas	Fresh	\$ 0.52	per pound	\$0.64	\$	0.33	pounds \$ 0.27
Berries, mixed	Frozen	\$ 3.56	per pound	\$1.00	\$	0.33	pounds \$ 1.18
Blackberries	Fresh	\$ 6.02	per pound	\$0.96	\$	0.32	pounds \$ 2.00
Blackberries	Frozen	\$ 3.64	per pound	\$1.00	\$	0.33	pounds \$ 1.20
Blueberries	Fresh	\$ 4.17	per pound	\$0.95	\$	0.32	pounds \$ 1.40
Blueberries	Frozen	\$ 3.39	per pound	\$1.00	\$	0.33	pounds \$ 1.12
Cantaloupe	Fresh	\$ 0.58	per pound	\$0.51	\$	0.37	pounds \$ 0.42
Cherries	Fresh	\$ 3.43	per pound	\$0.92	\$	0.34	pounds \$ 1.27
Cherries, packed in syrup or water	Canned	\$ 4.53	per pound	\$0.65	\$	0.44	pounds \$ 3.07
Clementines	Fresh	\$ 1.38	per pound	\$0.77	\$	0.46	pounds \$ 0.83
Cranberries	Dried	\$ 4.65	per pound	\$1.00	\$	0.12	pounds \$ 0.57
Dates	Dried	\$ 5.57	per pound	\$1.00	\$	0.17	pounds \$ 0.92
Figs	Dried	\$ 6.84	per pound	\$0.96	\$	0.17	pounds \$ 1.18
Fruit cocktail, packed in juice	Canned	\$ 1.72	per pound	\$1.00	\$	0.54	pounds \$ 0.93
Fruit cocktail, packed in syrup or water	Canned	\$ 1.59	per pound	\$0.65	\$	0.44	pounds \$ 1.08
Grapefruit	Fresh	\$ 1.17	per pound	\$0.49	\$	0.46	pounds \$ 1.11
Grapefruit, ready-to-drink	Juice	\$ 1.04	per pint	\$1.00	\$	8.00	fluid ounces \$ 0.52
Grapes	Fresh	\$ 1.84	per pound	\$0.96	\$	0.33	pounds \$ 0.63
Grapes (raisins)	Dried	\$ 3.78	per pound	\$1.00	\$	0.17	pounds \$ 0.63
Grapes, ready-to-drink	Juice	\$ 0.92	per pint	\$1.00	\$	8.00	fluid ounces \$ 0.46
Grapes, frozen concentrate	Juice	\$ 0.71	per pint	\$1.00	\$	8.00	fluid ounces \$ 0.36
Honeydew	Fresh	\$ 0.91	per pound	\$0.46	\$	0.37	pounds \$ 0.74
Kiwi	Fresh	\$ 2.18	per pound	\$0.76	\$	0.39	pounds \$ 1.11
Mangoes	Fresh	\$ 1.15	per pound	\$0.71	\$	0.36	pounds \$ 0.59
Mangoes	Dried	\$ 10.55	per pound	\$1.00	\$	0.13	pounds \$ 1.32
Nectarines	Fresh	\$ 1.91	per pound	\$0.91	\$	0.32	pounds \$ 0.67
Oranges	Fresh	\$ 1.21	per pound	\$0.68	\$	0.41	pounds \$ 0.73
Oranges, ready-to-drink	Juice	\$ 0.96	per pint	\$1.00	\$	8.00	fluid ounces \$ 0.49
Oranges, frozen concentrate	Juice	\$ 0.77	per pint	\$1.00	\$	8.00	fluid ounces \$ 0.38
Papaya	Fresh	\$ 1.29	per pound	\$0.62	\$	0.31	pounds \$ 0.64
Papaya	Dried	\$ 5.51	per pound	\$1.00	\$	0.15	pounds \$ 0.85
Peaches	Fresh	\$ 1.72	per pound	\$0.96	\$	0.34	pounds \$ 0.61
Peaches, packed in juice	Canned	\$ 2.02	per pound	\$1.00	\$	0.54	pounds \$ 1.09
Peaches, packed in syrup or water	Canned	\$ 1.81	per pound	\$0.65	\$	0.44	pounds \$ 1.23
Peaches	Frozen	\$ 3.39	per pound	\$1.00	\$	0.33	pounds \$ 1.12
Pears	Fresh	\$ 1.59	per pound	\$0.90	\$	0.36	pounds \$ 0.64
Pears, packed in juice	Canned	\$ 1.95	per pound	\$1.00	\$	0.54	pounds \$ 1.06
Pears, packed in syrup or water	Canned	\$ 1.90	per pound	\$0.65	\$	0.44	pounds \$ 1.29
Pineapple	Fresh	\$ 0.57	per pound	\$0.51	\$	0.36	pounds \$ 0.41
Pineapple, packed in juice	Canned	\$ 1.43	per pound	\$1.00	\$	0.54	pounds \$ 0.77
Pineapple, packed in syrup or water	Canned	\$ 1.41	per pound	\$0.65	\$	0.44	pounds \$ 0.95
Pineapple	Dried	\$ 6.65	per pound	\$1.00	\$	0.15	pounds \$ 1.03
Pineapple, ready-to-drink	Juice	\$ 1.03	per pint	\$1.00	\$	8.00	fluid ounces \$ 0.51
Pineapple, frozen concentrate	Juice	\$ 0.70	per pint	\$1.00	\$	8.00	fluid ounces \$ 0.35
Plum	Fresh	\$ 2.03	per pound	\$0.94	\$	0.36	pounds \$ 0.79
Plum (prunes)	Dried	\$ 5.70	per pound	\$1.00	\$	0.19	pounds \$ 1.07
Plum (prune), ready-to-drink	Juice	\$ 1.55	per pint	\$1.00	\$	8.00	fluid ounces \$ 0.78
Pomegranate	Fresh	\$ 2.24	per pound	\$0.56	\$	0.34	pounds \$ 1.36
Pomegranate, ready-to-drink	Juice	\$ 3.12	per pint	\$1.00	\$	8.00	fluid ounces \$ 1.56
Raspberries	Fresh	\$ 6.64	per pound	\$0.96	\$	0.32	pounds \$ 2.21
Raspberries	Frozen	\$ 4.19	per pound	\$1.00	\$	0.33	pounds \$ 1.38
Strawberries	Fresh	\$ 2.58	per pound	\$0.94	\$	0.32	pounds \$ 0.88
Strawberries	Frozen	\$ 2.82	per pound	\$1.00	\$	0.33	pounds \$ 0.93
Watermelon	Fresh	\$ 0.36	per pound	\$0.52	\$	0.33	pounds \$ 0.23

MEMBER ID
Job 1

References Add-ins Excel Add-ins Button Group Box

Macro name: Nationals_Baby

Macros in: This Workbook

Description: This macro will format tables

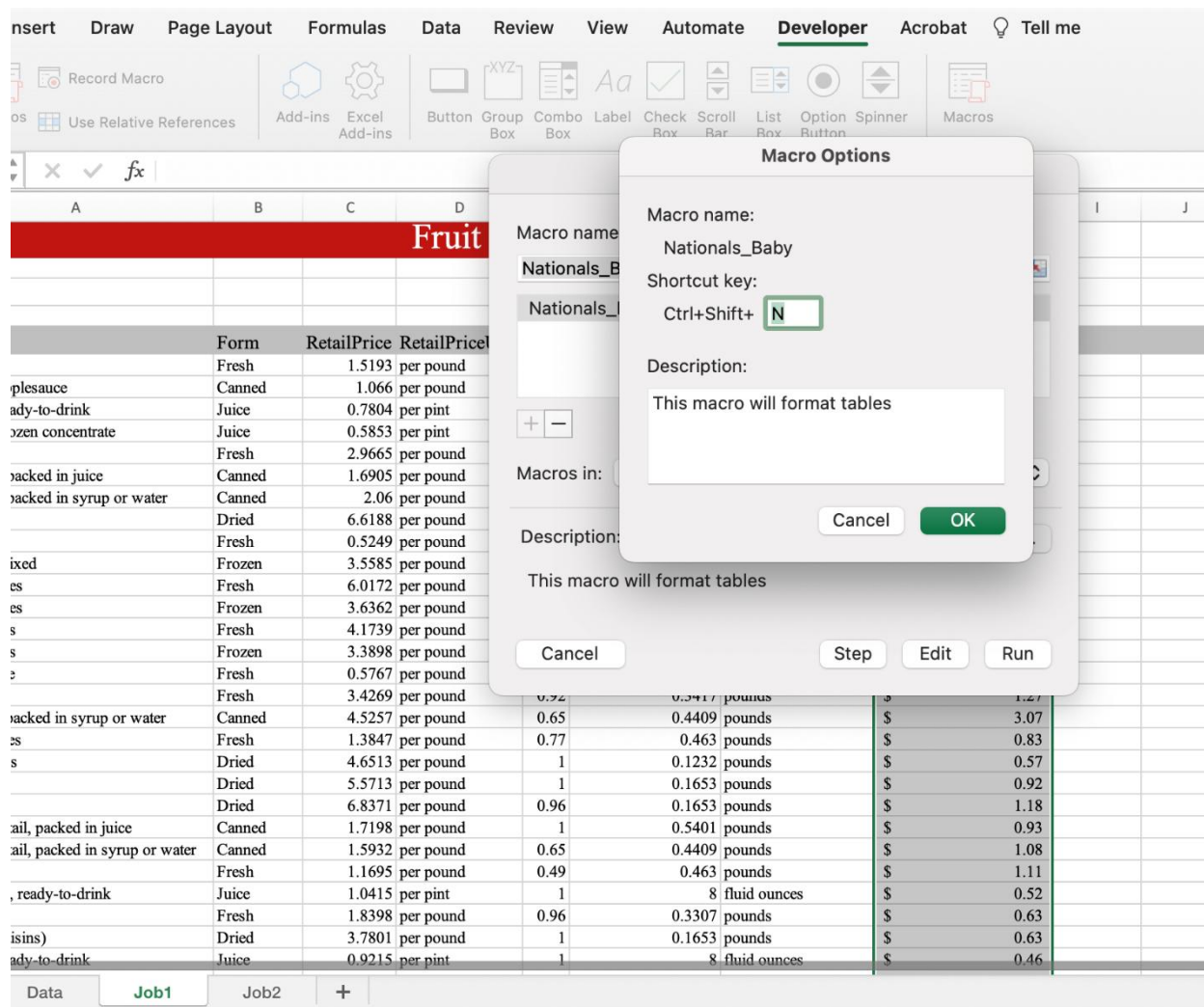
Cancel OK

Form	RetailPrice	RetailPriceUnit
Fresh	\$ 1.52	per pound
Canned	\$ 1.07	per pound
Juice	\$ 0.78	per pint
Juice	\$ 0.59	per pint
Fresh	\$ 2.97	per pound
Canned	\$ 1.69	per pound
Canned	\$ 2.06	per pound
Dried	\$ 6.62	per pound
Fresh	\$ 0.52	per pound
Frozen	\$ 3.56	per pound
Fresh	\$ 6.02	per pound
Frozen	\$ 3.64	per pound
Fresh	\$ 4.17	per pound
Frozen	\$ 3.39	per pound
Fresh	\$ 0.58	per pound
Fresh	\$ 3.43	per pound
Canned	\$ 4.53	per pound
Fresh	\$ 1.38	per pound
Dried	\$ 4.65	per pound
Dried	\$ 5.57	per pound
Dried	\$ 6.84	per pound
Canned	\$ 1.72	per pound
Canned	\$ 1.59	per pound
Fresh	\$ 1.17	per pound
Juice	\$ 1.04	per pint
Fresh	\$ 1.84	per pound
Dried	\$ 3.78	per pound
Juice	\$ 0.92	per pint

Job2 +

Average: \$1.53 Count: 505 Sum: \$379.25

Right-click the shape and select Assign Macro.



Macro Visual Basic code:

Sub Nationals_Baby()

,

' Nationals_Baby Macro

' This macro will format tables

,

' Keyboard Shortcut: Ctrl+Shift+N

,

Cells.Select

Rows("2:2").Select

```
Rows("2:4").Select
```

```
Selection.Insert Shift:=xlDown, CopyOrigin:=xlFormatFromLeftOrAbove
```

```
Rows("1:1").Select
```

```
Selection.RowHeight = 33
```

```
With Selection.Font
```

```
.Name = "Times New Roman"
```

```
.Size = 26
```

```
.Strikethrough = False
```

```
.Superscript = False
```

```
.Subscript = False
```

```
.OutlineFont = False
```

```
.Shadow = False
```

```
.Underline = xlUnderlineStyleNone
```

```
.ThemeColor = xlThemeColorLight1
```

```
.TintAndShade = 0
```

```
.ThemeFont = xlThemeFontNone
```

```
End With
```

```
Range("A1:H1").Select
```

```
With Selection
```

```
.HorizontalAlignment = xlCenter
```

```
.VerticalAlignment = xlBottom
```

```
.WrapText = False
```

```
.Orientation = 0
```

```
.AddIndent = False
```

```
.IndentLevel = 0
```

```
.ShrinkToFit = False
```

```
.ReadingOrder = xlContext
```

```
.MergeCells = False
```

```
End With
```

Selection.Merge

With Selection.Interior

.Pattern = xlSolid

.PatternColorIndex = xlAutomatic

.Color = 192

.TintAndShade = 0

.PatternTintAndShade = 0

End With

Range("A1:H1").Select

With Selection.Font

.ThemeColor = xlThemeColorDark1

.TintAndShade = 0

End With

Range("2:2,4:4").Select

Range("A4").Activate

Selection.RowHeight = 15

Rows("3:3").Select

Selection.RowHeight = 20

Range("A5:H5").Select

Selection.RowHeight = 20

With Selection.Font

.Name = "Times New Roman"

.Size = 14

.Strikethrough = False

.Superscript = False

.Subscript = False

.OutlineFont = False

.Shadow = False

.Underline = xlUnderlineStyleNone

```
.ThemeColor = xlThemeColorLight1
```

```
.TintAndShade = 0
```

```
.ThemeFont = xlThemeFontNone
```

```
End With
```

```
With Selection.Interior
```

```
.Pattern = xlSolid
```

```
.PatternColorIndex = xlAutomatic
```

```
.ThemeColor = xlThemeColorDark1
```

```
.TintAndShade = -0.249977111117893
```

```
.PatternTintAndShade = 0
```

```
End With
```

```
With Selection.Interior
```

```
.Pattern = xlSolid
```

```
.PatternColorIndex = xlAutomatic
```

```
.ThemeColor = xlThemeColorDark1
```

```
.TintAndShade = -0.249977111117893
```

```
.PatternTintAndShade = 0
```

```
End With
```

```
Columns("H:H").Select
```

```
Range("H2").Activate
```

```
Selection.Style = "Currency"
```

```
End Sub
```

Job 2 Print Out

Fruit Prices						
Select Form:		Juice				
Fruit	Form	RetailPrice	RetailPriceUnit	Yield	CupEquivalentSize	CupEquivalentPrice
Apples	Fresh	1.5193	per pound	0.9	0.2425 pounds	\$ 0.41
Apples, applesauce	Canned	1.066	per pound	1	0.5401 pounds	\$ 0.58
Apples, ready-to-drink	Juice	0.7804	per pint	1	8 fluid ounces	\$ 0.39
Apples, frozen concentrate	Juice	0.5853	per pint	1	8 fluid ounces	\$ 0.29
Apricots	Fresh	2.9665	per pound	0.95	0.3638 pounds	\$ 1.16
Apricots, packed in juice	Canned	1.6905	per pound	1	0.5401 pounds	\$ 0.91
Apricots, packed in syrup or water	Canned	2.06	per pound	0.65	0.4409 pounds	\$ 1.40
Apricots	Dried	6.6168	per pound	1	0.1433 pounds	\$ 0.95
Bananas	Fresh	0.5249	per pound	0.64	0.3307 pounds	\$ 0.27
Berries, mixed	Frozen	3.5585	per pound	1	0.3307 pounds	\$ 1.18
Blackberries	Fresh	6.0172	per pound	0.96	0.3197 pounds	\$ 2.00
Blackberries	Frozen	3.6361	per pound	1	0.3307 pounds	\$ 1.20
Blueberries	Fresh	4.1739	per pound	0.95	0.3197 pounds	\$ 1.40
Blueberries	Frozen	3.3898	per pound	1	0.3307 pounds	\$ 1.12
Cantaloupe	Fresh	0.5787	per pound	0.51	0.3748 pounds	\$ 0.42
Cherries	Fresh	3.4269	per pound	0.92	0.3417 pounds	\$ 1.27
Cherries, packed in syrup or water	Canned	4.5257	per pound	0.65	0.4409 pounds	\$ 3.07
Clementines	Fresh	1.3847	per pound	0.77	0.463 pounds	\$ 0.83
Cranberries	Dried	4.6513	per pound	1	0.1232 pounds	\$ 0.57
Dates	Dried	5.5713	per pound	1	0.1653 pounds	\$ 0.92
Figs	Dried	6.8371	per pound	0.96	0.1653 pounds	\$ 1.18
Fruit cocktail, packed in juice	Canned	1.7158	per pound	1	0.5401 pounds	\$ 0.93
Fruit cocktail, packed in syrup or water	Canned	1.5952	per pound	0.65	0.4409 pounds	\$ 1.08
Grapefruit	Fresh	1.1695	per pound	0.49	0.463 pounds	\$ 1.11
Grapefruit, ready-to-drink	Juice	1.0415	per pint	1	8 fluid ounces	\$ 0.52
Grapes	Fresh	1.8398	per pound	0.96	0.3307 pounds	\$ 0.63
Grapes (raisins)	Dried	3.7801	per pound	1	0.1653 pounds	\$ 0.63
Grapes, ready-to-drink	Juice	0.9215	per pint	1	8 fluid ounces	\$ 0.46
Grapes, frozen concentrate	Juice	0.7118	per pint	1	8 fluid ounces	\$ 0.36
Honeydew	Fresh	0.9056	per pound	0.46	0.3748 pounds	\$ 0.74
Kiwi	Fresh	2.1849	per pound	0.76	0.3858 pounds	\$ 1.11
Mangoes	Fresh	1.1513	per pound	0.71	0.3638 pounds	\$ 0.59
Mangoes	Dried	10.5527	per pound	1	0.1253 pounds	\$ 1.32
Nectarines	Fresh	1.9062	per pound	0.91	0.3197 pounds	\$ 0.67
Oranges	Fresh	1.2131	per pound	0.68	0.4079 pounds	\$ 0.73
Oranges, ready-to-drink	Juice	0.9042	per pint	1	8 fluid ounces	\$ 0.49
Oranges, frozen concentrate	Juice	0.769	per pint	1	8 fluid ounces	\$ 0.38
Papaya	Fresh	1.2904	per pound	0.62	0.3086 pounds	\$ 0.64
Papaya	Dried	5.5089	per pound	1	0.1543 pounds	\$ 0.85
Peaches	Fresh	1.7167	per pound	0.96	0.3417 pounds	\$ 0.61
Peaches, packed in juice	Canned	2.0237	per pound	1	0.5401 pounds	\$ 1.09
Peaches, packed in syrup or water	Canned	1.8117	per pound	0.65	0.4409 pounds	\$ 1.23
Peaches	Frozen	3.3867	per pound	1	0.3307 pounds	\$ 1.12
Pears	Fresh	1.5865	per pound	0.9	0.3638 pounds	\$ 0.64
Pears, packed in juice	Canned	1.9546	per pound	1	0.5401 pounds	\$ 1.06
Pears, packed in syrup or water	Canned	1.897	per pound	0.65	0.4409 pounds	\$ 1.29
Pineapple	Fresh	0.5685	per pound	0.51	0.3638 pounds	\$ 0.41
Pineapple, packed in juice	Canned	1.4344	per pound	1	0.5401 pounds	\$ 0.77
Pineapple, packed in syrup or water	Canned	1.4067	per pound	0.65	0.4409 pounds	\$ 0.95
Pineapple	Dried	6.6482	per pound	1	0.1543 pounds	\$ 1.03
Pineapple, ready-to-drink	Juice	1.0288	per pint	1	8 fluid ounces	\$ 0.51
Pineapple, frozen concentrate	Juice	0.6973	per pint	1	8 fluid ounces	\$ 0.35
Plum	Fresh	2.0292	per pound	0.94	0.3638 pounds	\$ 0.79
Plum (prunes)	Dried	5.7042	per pound	1	0.1874 pounds	\$ 1.07
Plum (prune), ready-to-drink	Juice	1.5522	per pint	1	8 fluid ounces	\$ 0.78
Pomegranate	Fresh	2.235	per pound	0.56	0.3417 pounds	\$ 1.36
Pomegranate, ready-to-drink	Juice	1.125	per pint	1	8 fluid ounces	\$ 0.56
Raspberries	Fresh	6.6391	per pound	0.96	0.3197 pounds	\$ 2.21
Raspberries	Frozen	4.1877	per pound	1	0.3307 pounds	\$ 1.38
Strawberries	Fresh	2.58	per pound	0.94	0.3197 pounds	\$ 0.88
Strawberries	Frozen	2.8189	per pound	1	0.3307 pounds	\$ 0.93
Watermelon	Fresh	0.3604	per pound	0.52	0.3307 pounds	\$ 0.23

Job 3 Print Out

Salary by Gender & Ehtnicity		
SEX	F	
	Data	
ETHNICITY	Sum of ANNUAL SALAI	Count of DEPARTMENT
American Indian/Alaskan Native	\$ 100,559.47	2
Asian or Pacific Islander	\$ 2,612,067.93	53
Black	\$ 32,416,183.12	645
Hispanic	\$ 1,183,663.11	26
Two or More Races	\$ 1,321,874.54	26
White (Not of Hispanic Origin)	\$ 122,903,315.13	2341
Grand Total	\$ 160,537,663.30	3093

MEMBER ID
Job 3